

Example 15.2

$$X_t = Y_t - Y_{t-1}$$

$$= \sum_{j=-\infty}^{\infty} b_j Y_{t-j} \quad \text{where}$$

$$b_0 = 1;$$

$$~~b_{-1} = -1~~ \quad b_1 = -1;$$

$$b_j = 0 \text{ otherwise.}$$

The transfer function is

$$h(\omega) = \sum_{j=-\infty}^{\infty} b_j e^{-i\omega j}$$

$$= 1 - e^{-i\omega}$$

$$g_X(\omega) = |h(\omega)|^2 g_Y(\omega)$$

$$|h(\omega)|^2 = (1 - e^{-i\omega})(1 - e^{-i\omega})$$

$$= (1 - e^{-i\omega})(1 - e^{i\omega})$$

$$= 2 - (e^{i\omega} + e^{-i\omega})$$

$$= 2 - 2\cos\omega$$

$$= 2(1 - \cos\omega)$$

$$|z|^2 = z\bar{z}$$

$$\therefore g_X(\omega) = \underline{2(1 - \cos\omega)} g_Y(\omega).$$